Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000
COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: **Bachelor of Science in Geodetic Engineering**

Primary Job Roles students are prepared for in the program	**Land Surveyor, Mine Surveyor, Geodesist, Photogrammetrist, Cartographer, GIS Specialist, Remote Sensing Specialist, Hydrographer**
Secondary Job Roles students are prepared for in the program	**Academician, Consultancy, Government Technical Service, Quality Assurance Engineer, Project Manager, Assessor, Realtor, Appraiser**

Primary and Secondary Job Roles provide a mental anchor to the identification of program outcomes you will be having for your program. This will allow you to check for completeness in your program outcomes.

Current CMO Guiding Program: CMO 89, Series of 2017

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "Higher Education Act of 1994," in pursuance of policies, standards and guidelines in the establishment of an outcomes-based education (OBE) system in higher education institutions offering engineering programs as advocated under CMO 37 s. 2012 and policy-standard to enhance quality assurance (QA) in Philippine higher education through an outcomes-based and typology-based QA as advocated under CMO 46 s. 2012, and by virtue of Commission en banc Resolution No. 788-2017 dated October 24, 2017 the following policies and standards are hereby adopted and promulgated by the Commission.

This CMO integrates the establishment of an outcomes-based Based on the Guidelines for the Implementation of CMO No. 46 series of 2012 and CMO 37 s. 2012, this PSG implements the shift to outcomes-based education (OBE) leading to competency-based standards. It specifies the "core competencies" expected of Bachelor of Science in Geodetic Engineering (BSGE) graduates "regardless of the type of Higher Education Institutions (HEI) they graduate from." However, in recognition of outcomes-based education and the typology of HEIs, this PSG also provide ample space for HEIs to innovate in the curriculum in line with the assessment of how best to achieve learning outcomes in their particular contexts and their respective missions.

Program Description (CMO 89, Series of 2017):

Definition
GEODETIC ENGINEERING is that branch of engineering which deals with the collection and measurement of spatial data above, on, or below the surface of the earth using appropriate technologies and the scientific and methodological processing and management of these data for the production of spatial information systems, maps, plans, charts, and other documents.

Degree Name
In consonance with the provisions stipulated in Section 2, Article II of RA 8560 also known as the Philippine Geodetic Engineering Act of 1998, as amended by RA 9200, the degree program herein shall be called BACHELOR OF SCIENCE IN GEODETIC ENGINEERING (BSGE).

Nature & Scope of Practice of a GE Graduate
The practice of Geodetic Engineering is a professional and organized act of gathering physical data on the surface of the earth with the use of precision instruments. It is also the scientific and methodical processing of these data and presenting them on graphs, plans, maps, charts or documents. It shall embrace, but is not limited to the following activities: Professional Geodetic Engineering services with the use of surveying and mapping equipment such as graduated rods, measuring tapes, transits, levels, theodolites, fathometers/echo sounders, electronic distance meters, global positioning systems, stereoplotters and other instruments that are used to determine metes and bounds of lands positions of points on the surface of the earth, water depths, underwater configuration, ground elevation, gravity, isostasy, crustal movements and the size and shape of the earth, and other instruments used for construction survey, and those instruments used to guide the installation of large industrial equipment and machineries; Horizontal and vertical control surveys and political boundary surveys; Land surveys to determine their metes and bounds and prepare the plans thereof for titling and for other purposes; Subdivision, consolidation and/or consolidation-subdivision of titled properties; Submission of survey plans of subdivided, consolidated and/or consolidated subdivision titled properties to the government agencies concerned; using the numbering system and projection prescribed by the government; Preparation and making of sketch, lot and location plans; Conduction of engineering surveys and the technical preparation of engineering survey plans such as topographic, hydrographic, tidal, profile, cross-section, construction and boundary surveys; Parcellary surveys of lands traversed by infrastructure projects; and the preparation of subdivision plans; Conduction of gravimetric and photogrammetric survey and the technical preparation of such survey plans; Survey and mapping works such as the preparation of geographic and/or land information systems; Survey to determine and establish line and grade for the construction of buildings and other structures and its attachments; Construction of as-staked and as-built surveys for infrastructures; Conduction of mineral and mining surveys; Installation of machineries requiring the use of precision instruments; Engagement in the transfer of the knowledge and technology of geodetic engineering in any institution of learning; Provision of consultancy services in geodetic engineering.

VISION
A PREMIER UNIVERSITY IN THE ASEAN REGION

MISSION
USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management.

CORE VALUES
USeP is a community of scholars that values:
U – Unity
S – Stewardship
E – Excellence
P – Professionalism

IGA4 IGA5 IGA6	f	Understand professional, social and ethical responsibility	1	Internalizes the professional code of ethics for specific discipline.
			2	Applies the principles of professional ethical standard and social responsibility to actual practice
IGA4	g	Communicate effectively	1	Articulate the convention and style in oral communication that conveys complete thought and information.
IGA5			2	Articulate the convention and style in written communication that conveys complete thought and information.
IGA6			3	Demonstrate proficiency in using graphs, tables, drawings, symbols, etc. applicable in the discipline.
IGA1, IGA3 IGA4, IGA6	h	Understand the impact of engineering solutions in a global, economic, environmental, & societal context through broad education	1	Relate engineering solutions to global, economic, environmental, and societal context.
			2	Describe engineering solutions in application to global, economic, environmental, and societal context.
IGA1, IGA2 IGA3, IGA4 IGA5, IGA6	i	Recognize the need for, and an ability to engage in life-long learning	1	Demonstrate ability to learn independently.
			2	Shows appreciation and understanding of the engineering profession.
			3	Understand the linkage or connection between his/her task to previous, intervening and subsequent tasks.
IGA2, IGA3 IGA4, IGA5 IGA6	j	Apply knowledge of contemporary issues	1	Identify the socio-economic, political, and environmental issues related to the practice of engineering
			2	Explain the possible impact of technology to local, national, and international issues
			3	Shows leadership skills in managing a multidisciplinary group.
IGA1 IGA4 IGA6	k	Use techniques, skills, and modern engineering tools necessary for engineering practice	1	Handles the details of the work assigned to him/her.
			2	Demonstrate the ability to operate the equipment needed in the engineering practice.
			3	Use modern technology in performing engineering practice.
IGA1, IGA2 IGA3, IGA4 IGA5, IGA6	l	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments	1	Capable of learning from and listening to co-members.
			2	Shows leadership skills in managing a multidisciplinary group.
			3	Participates in discussions and comes up with sound suggestions to problems.

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome.

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CURRICULUM MAP

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

Year Level	Semester	Units	Courses in the Curriculum	Program Outcomes											
				a	b	c	d	e	f	g	h	i	j	k	l
FIRST YEAR	FIRST	3	Purposive Communication												
		3	Mathematics in the Modern World												
		1	Geodetic Engineering Orientation				I		I				I		
		3	General Surveying 1	I				I				I		I	
		1	Engineering Drawing and Plans			I				I					
		3	Computer Fundamentals and Programming			I				I					
		3	Engineering Calculus 1	I											
		2	Movement Enhancement												
		3	National Service Training Program 1												
	SECOND	4	General Surveying 2	I				I				I		I	
		2	Intro to the Laws on Private & Public Lands						I	I			I		
		4	Physics for Engineers	I	I										
		2	Computer-Aided Drafting			I								I	
		3	Principles of Geology	I									I		
		3	Engineering Calculus 2	I											
		2	Fitness Exercise												
		3	National Service Training Program 1												
OFF SEMESTER		3	Differential Equations	I											
		3	Engineering Mechanics	I											
SECOND YEAR	FIRST	4	Engineering Surveys	E				E						E	
		3	Theory of Errors & Adjustments	I				I							
		3	Geometric Geodesy	E				E							
		4	Cartography & Map Projections			I				I				I	
		3	Geodetic Astronomy	E				E							
		3	Engineering Data Analysis	I											
		3	Advanced Information & Communications Tech.			I				I					
		2	Physical Activities Towards Health & Fitness I												
	SECOND	3	Understanding the Self												
		3	Science, Technology, and Society												
		3	The Life and Works of Rizal												
		3	Readings in Philippine History												
		3	Hydrographic Surveying	E				E						E	
		3	Physical Geodesy	E				E						E	

		2	GE Laws, Obligations & Contracts, & Ethics						E	E			E		
		3	Public Land Laws & Laws on Natural Resources						E	E			E		
		2	Physical Activities Towards Health & Fitness I												
OFF SEMESTER		1	Survey Camp - 120 Hours	E				E						E	
THIRD	FIRST	3	General Education Elective 1												
		3	Satellite Geodesy	E				E						E	
		4	Geodetic Computations & Adjustments	E				E						E	
		3	Land Registration Laws						E	E			E		
		3	Geographic Information Systems	E		E		E				E		E	
		2	Mine Surveying	E			E							E	
		3	Engineering Economics	E									E		
		3	Electrical & Electronics Eng'g. for Geodetic Engrs.	I											
	SECOND	3	General Education Elective 2												
		4	Geodetic Surveying	D				D						D	
		4	Photogrammetry	E				E						E	
		5	Property Surveys					E	E					E	
		3	Methods of Research		E	E				E					
		3	Engineering Management	E											E
		3	Technopreneurship												
OFF SEMESTER		2	GE Immersion/OJT - (320 Hours)	D			D	D	D	D	D	D		D	D
FOURTH	FIRST	3	General Education Elective 3												
		3	Art Appreciation												
		3	Contemporary World												
		3	Special Studies in Geodetic Engineering		D	D		D		D	D	D		D	
		4	Remote Sensing	D		E	E	E	E		E				E
		3	Land Administration & Management					D	D	D	D		D		
		3	GEn Elective				D				D				
		3	Intro to Environmental Science & Engineering				E						E		
	SECOND	3	Ethics												
		3	Special Studies in Geodetic Engineering		D	D		D		D	D	D		D	
		3	Land Use Planning & Development			D	D	D	D		D				D
		2	Correlation Course for GE	D				D							
		1	Free Elective							E			E		
		3	Basic Occupational Health & Safety												

Program Educational Objectives		Mission-Vision and Core Value Alignment						Reference Source or CMO Provision Compliance
		Vision	Mission	Core Value				
				U	S	E	P	
PEO1	Demonstrate expertise in the field of Geodetic Engineering	√	√		√	√	√	USEP Brochure, CMO 89, s. 2017
PEO2	Actively participate in professional communities to nourish ever-developing careers	√	√	√	√		√	USEP Brochure, CMO 89, s. 2017
PEO3	Demonstrate gender-sensitive leadership and commitment to work in diverse cultural and interdisciplinary backgrounds	√	√	√		√	√	USEP Brochure, CMO 89, s. 2017
PEO4	Exhibit globally competitive skills	√	√		√	√	√	USEP Brochure, CMO 89, s. 2017

PERFORMANCE INDICATORS

Complete the performance indicators for each of the outcomes that we are able to generate during the workshop. The starting statements are not yet complete; they are just starting phrases to guide you in your performance indicators formulation.

Institutional Graduate Attributes:

IGA1 – Critical Analysis; IGA2 – Leadership and Teamwork; IGA3 – Service-Oriented; IGA4 – Professional Competence; IGA5 – Communication; IGA6 – Life-Long Learning

Program Attribute		Program Outcomes		Performance Indicators
IGA1 IGA4 IGA6	a	Apply knowledge of mathematics and sciences to solve engineering problems;	1 2 3	Apply concepts in mathematics to solve engineering problems Apply principles in chemical sciences to solve engineering problems. Apply principles in physical sciences to solve engineering problems
IGA1, IGA4 IGA5, IGA6	b	Design and conduct experiments, as well as to analyze and interpret data	1 2	Perform laboratory experiments as prescribed by standard procedures. Design an experiment to achieve a specific objective
IGA1, IGA2 IGA3, IGA4 IGA5, IGA6	c	Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.	1 2	Develop an engineering solution that have various constraints as factors Design a system or process based on various constraints
IGA2 IGA3 IGA6	d	Function on multidisciplinary teams	1 2 3	Shows tact in dealing with people he/she comes in contact with. Capable of learning from and listening to co-members Shows respect and courtesy in dealing with peers and superiors.
IGA1 IGA4 IGA6	e	Identify, formulate, and solve engineering problems	1 2 3	Interpret statement problems into mathematical expressions. Apply concepts, theories, and principles in solving engineering problem(s) Solve specific engineering problem related to discipline.

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

5 Very Likely
4 Less Likely
3 Cannot Tell
2 Not Likely
1 Definitely Not

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	√				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	√				
Do you think that your course planning technique has gone better this time than before?	√				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	√				
Do you think your topics will be better anchored to a specific purpose/s than before?	√				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	√				
Do you think that if this course will be properly implemented, your students will be developed better?	√				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	√				

Prepared:

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